Tejas Malhotra

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M.Sc. ICT Student | Software Engineering | Distributed Systems and Backend Development | Python and AI

EDUCATION

M.Sc. Information and Communication Technology *10/2025 - Present*
Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen, Germany

B.Tech. Electronics and Communication Engineering *09/2020 - 10/2024*
Manipal University Jaipur, Jaipur, India

PROJECTS

RoomLock - Concurrency-Safe Booking API - *Python | FastAPI | PostgreSQL | Docker | JWT | Locust* | GitHub | Live Demo

- Found and reproduced a booking race condition under concurrent load, then reasoned through why "check availability, then insert" fails, and eliminated it two different ways: a PostgreSQL SELECT FOR UPDATE row lock, and a GiST EXCLUDE constraint enforced at commit time.
- Load-tested both fixes with Locust (30 concurrent users), measuring throughput at each stage: ~420 req/s after row-level locking, ~450-570 req/s after adding the exclusion constraint, both eliminating all overlapping bookings (versus 6 overlapping pairs unfixed).
- Verified correctness with 66 automated tests (pytest) against a real PostgreSQL instance, including a concurrency test that races 20 identical requests and asserts exactly one succeeds.
- Designed the schema, API, and authentication (JWT, role-based access) end to end, then containerized the whole system with Docker for reproducible deployment.

ReviewLenzAI - AI-Powered GitHub PR Review Tool - *Python | FastAPI | React/TypeScript | PostgreSQL | Groq (Llama 3.3 70B)* | GitHub | Live Demo

- Built a FastAPI backend that fetches GitHub PR diffs and sends them to an LLM (Llama 3.3 70B via Groq's API) with a structured tool-call schema, validated by Pydantic so findings return as typed data instead of free-form text.
- Designed multi-user isolation with HttpOnly-cookie auth and Fernet-encrypted GitHub tokens, and deployed the full stack independently: React/TypeScript frontend on Vercel, FastAPI backend on Render, PostgreSQL on Neon.
- Wrote 31 backend tests (pytest) wired into GitHub Actions CI on every push.

PROFESSIONAL EXPERIENCE

Cloud Intern | Parity Infotech Solutions, Bangalore, India *01/2024 - 05/2024*

- Contributed to AWS infrastructure (EC2, VPC, RDS, S3) supporting production workloads, applying reliability improvements to reduce downtime.
- Built and maintained GitHub Actions and GitLab CI/CD pipelines automating build, test, and release workflows.

IoT Intern | Parity Infotech Solutions, New Delhi, India *05/2023 - 08/2023*

- Built Python data-integration pipelines for sensor and IoT traffic data supporting smart-city applications.
- Diagnosed and resolved 20+ performance-critical defects through systematic analysis of application logs and system metrics, tracing each back to root cause.

TECHNICAL SKILLS

Programming and CS Fundamentals: Python, object-oriented design, data structures, algorithms, complexity analysis

Databases and Distributed Systems: PostgreSQL, MySQL, transactions, isolation levels, locking, concurrency control, relational schema design

Backend and APIs: FastAPI, REST APIs, Pydantic

AI and GenAI Tooling: LLM API integration (Groq / Llama 3.3 70B) with schema-validated structured output; GitHub Copilot (daily use)

Cloud and DevOps: AWS (EC2, S3, RDS, VPC), Docker, CI/CD (GitHub Actions, GitLab CI/CD), Git

AWARDS AND LANGUAGES

IEEE MTT-S Winner | English: C2 | German: A2 (actively studying)